DEEPFAKE DETECTION

Test Plan & Test Cases

Test Strategy, Cases, and Acceptance Report

Major Prabhat Pandey | April 2026 | MLOps Project

1. Acceptance Criteria

Criterion	Target	Status
Model Accuracy	≥ 90% on held-out test set	PASS
F1-Score	≥ 0.90	PASS
P95 Inference Latency	< 200ms over 100 requests	PASS
API Availability	All endpoints return expected status codes	PASS
Pipeline Reproducibility	dvc repro produces identical artifacts	PASS

2. Test Strategy

Tests are organized into three tiers:

- **Unit Tests**: Individual functions tested in isolation using pytest and unittest.mock. Covers preprocessing, model forward pass, drift detection, and API response models.
- **Integration Tests**: FastAPI endpoints tested via TestClient with a real (or mocked) model. Covers all HTTP status codes and response schemas.
- **Acceptance Tests**: End-to-end performance validation — F1 ≥ 0.90 on test split, p95 latency < 2000ms on 100 requests.

3. Test Cases

Total test cases: 34 | Passing: 34 | Failed: 0

3.1 Unit Tests — Preprocessing

ID	Description	Expected Result	Status
TC-01	extract_frames returns list ≤ requested count from mocked video	List of numpy arrays, len == num_frames	PASS
TC-02	extract_frames raises ValueError when video cannot be opened	ValueError with 'Cannot open video'	PASS
TC-03	extract_frames raises ValueError when total_frames <= 0	ValueError with 'Video has no frames'	PASS
TC-04	detect_faces fallback returns (224,224,3) when no face detected	Shape == (224,224,3)	PASS
TC-05	preprocess_video output shape is (30, 3, 224, 224)	Tensor shape matches	PASS

3.2 Unit Tests — Model

ID	Description	Expected Result	Status
TC-06	DeepfakeDetector output shape (2,30,3,224,224) → (2,1)	Shape == (2,1)	PASS
TC-07	DeepfakeDetector output in [0,1]	All values in sigmoid range	PASS
TC-08	DeepfakeDetector gradients flow through all parameters	No frozen gradients	PASS

3.3 Unit Tests — Drift Detection

ID	Description	Expected Result	Status
TC-09	compute_drift_score returns < 1.0 at baseline mean	Score < 1.0	PASS
TC-10	compute_drift_score returns > 3.0 far from mean	Score > 3.0	PASS
TC-11	is_drifted returns True above threshold, False below	Boolean correct	PASS

3.4 Unit Tests — API Response Models

ID	Description	Expected Result	Status
TC-12	PredictResponse rejects invalid confidence (1.5)	ValidationError raised	PASS
TC-13	PredictResponse rejects invalid prediction ('unknown')	ValidationError raised	PASS
TC-24	POST /feedback with valid body returns {'status': 'logged'}	HTTP 200	PASS
TC-25	POST /feedback with invalid ground_truth ('unknown') returns 422	HTTP 422	PASS
TC-26	POST /admin/rollback with valid version triggers model reload	HTTP 200, model_version updated	PASS
TC-27	POST /admin/rollback when reload raises exception returns 500	HTTP 500	PASS
TC-28	GET /admin/model-info returns model_version, run_id, model_loaded	HTTP 200, all fields present	PASS

3.5 Unit Tests — Evaluation and Pipeline

ID	Description	Expected Result	Status
TC-29	evaluate.py extended metrics includes pr_auc, precision_fake, recall_fake	Dict contains all 7 keys	PASS
TC-30	find_best_threshold returns float in [0,1]	isinstance float, 0.0 <= t <= 1.0	PASS
TC-31	quantize_model saves a loadable file smaller than or equal to original	File exists, torch.load succeeds	PASS
TC-32	validate_feature_file returns errors for wrong-shape tensor	errors list non-empty	PASS
TC-33	validate_feature_file returns no errors for valid (30,3,224,224) tensor	errors list empty	PASS

3.6 Integration Tests

ID	Description	Expected Result	Status
TC-14	GET /health returns 200 with status='ok'	HTTP 200	PASS
TC-15	GET /ready returns 200 with model loaded	HTTP 200	PASS
TC-16	GET /ready returns 503 without model	HTTP 503	PASS
TC-17	GET /metrics returns Prometheus text	HTTP 200, body contains metric names	PASS
TC-18	POST /predict with .avi file returns 400	HTTP 400	PASS
TC-19	POST /predict with valid MP4 returns PredictResponse	HTTP 200, prediction in (real, fake)	PASS
TC-20	POST /predict confidence >= 0.5 gives prediction='fake'	prediction == 'fake'	PASS
TC-21	POST /predict confidence < 0.5 gives prediction='real'	prediction == 'real'	PASS

3.7 Acceptance Tests

ID	Description	Target	Status
TC-22	Model F1 ≥ 0.90 on test split	F1 >= 0.90	PASS
TC-23	P95 latency < 200ms over 100 requests	p95 < 200ms	PASS
TC-34	POST /predict with MP4 returns prediction in < 200ms (backend running)	HTTP 200, latency < 200ms	PASS

4. Test Report Summary

Category	Total	Passed	Failed
Unit — Preprocessing	5	5	0
Unit — Model	3	3	0
Unit — Drift Detection	3	3	0
Unit — API Models	7	7	0
Unit — Pipeline	5	5	0
Integration	8	8	0
Acceptance	3	3	0
TOTAL	34	34	0

All 34 test cases pass. Acceptance criteria are fully met.

5. Test Infrastructure

- Framework: pytest 8.x with pytest-asyncio for async endpoint tests
- HTTP testing: Starlette TestClient (wraps FastAPI app)
- Mocking: unittest.mock for cv2.VideoCapture and MTCNN in unit tests
- Run command: pytest tests/ -v (from deepfake-detection/)
- CI: dvc repro re-runs evaluate stage to validate F1 on every pipeline run